Newtonian Mechanics

Issac Newton (1642–1727) 1644: 李自成入京

1661: 康熙登基

- Relation on Force & Acceleration
- Appicability: $\begin{cases} \text{low speed } (v \ll c) & \rightarrow \text{relativity} \\ \text{macroscopic scale of interaction} & \rightarrow \text{quantum mechanics} \end{cases}$

Newton's First law:

If no force acts on a body, the body's velocity cannot change. (inertia)

$\underbrace{\text{velocity cannot change}}_{\text{no acceleration}}$

Background.

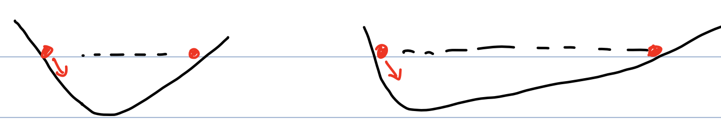
Aristotle (need continuous action of an agent)

(384BC – 322BC)

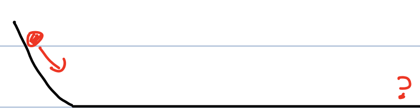
Galilei (objects retain their motion in absence of any impediments)

(1564–1642) (incorporated into Newton's 1st law)

Thought experiment:



Q:



- Force
  - contact force
  - field force (gravity, Electric, Magnetic, ...)

} (roughly speaking)

Newton's Second Law:

$$\vec{F} = m\vec{a}$$

The net force on a body is equal to the product of the body's mass and its acceleration.

$$F_x = ma_x, \quad F_y = ma_y, \quad F_z = ma_z$$

m : mass

Weight $\neq$ mass e.g. at space station $\rightarrow$ $\begin{cases} \text{weightless} \\ \text{mass same as on earth} \end{cases}$

Newton's Third Law:

$$\vec{F}_{12} = -\vec{F}_{21}$$

- equal magnitude
- opposite direction.

When two bodies interact, the forces on the bodies from each other are always equal in magnitude and opposite in direction.

Inertial reference frame :

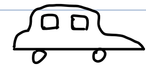
where Newton's Laws hold

Inertia: tendency to resist change.

Non-inertial frame:

. e.g. an accelerating car

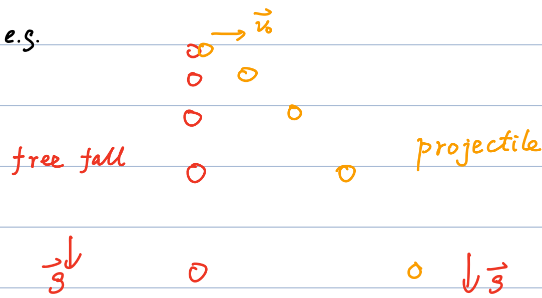
$\longrightarrow \vec{a}$



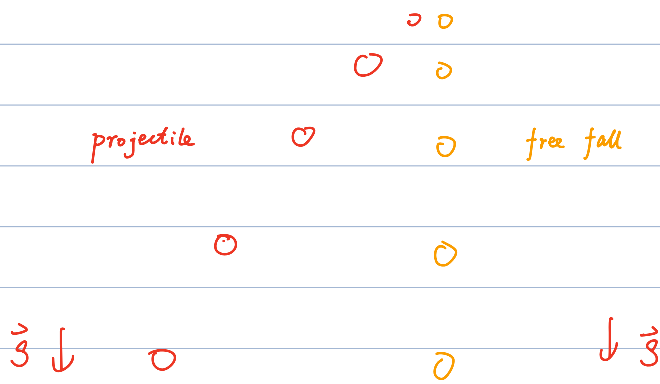
$\Big\langle$ pushed by seat / stationary.

Relative motion (inertial frame)

e.g.



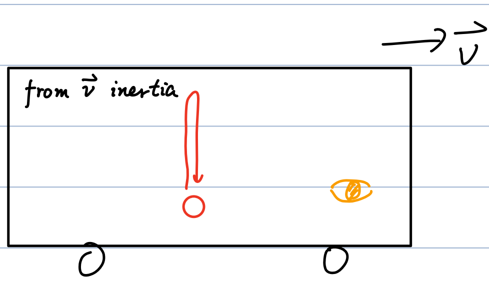
frame change to $\vec{v}_0$:

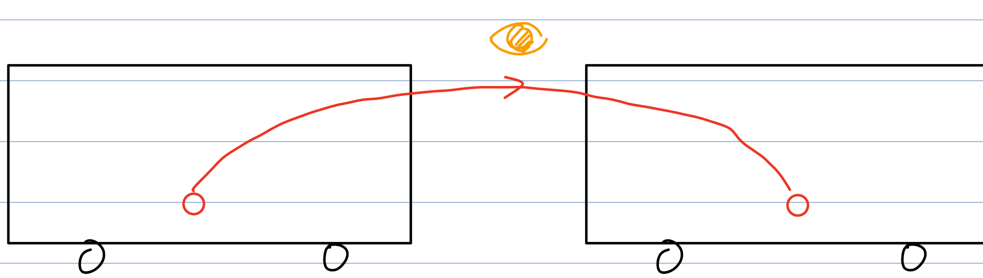


different inertia observers :

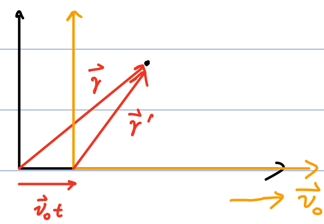
- agree on $\vec{a}$
- disagree on $\vec{x}$, $\vec{v}$

Relative motion (inertial frame)





Galilean transformation:



$\vec{v}_0$: t-independent

(From the plot)

$$\vec{r}\,' = \vec{r} - \vec{v}_0 t$$

$$\Rightarrow \frac{d\vec{r}\,'}{dt} = \frac{d\vec{r}}{dt} - \vec{v}_0$$

i.e. $\vec{v}\,' = \vec{v} - \vec{v}_0$

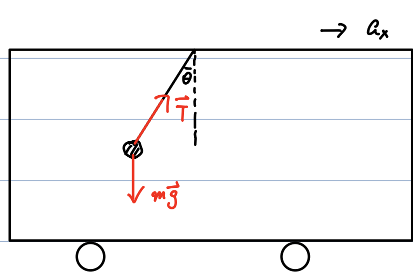
$$\Rightarrow \frac{d\vec{v}\,'}{dt} = \frac{d\vec{v}}{dt} \qquad \left(\frac{d\vec{v}_0}{dt} = 0\right)$$

i.e. $\vec{a}\,' = \vec{a}$

Relative motion (non-inertial frame)

(car moving with a_x)

(free-body diagram)

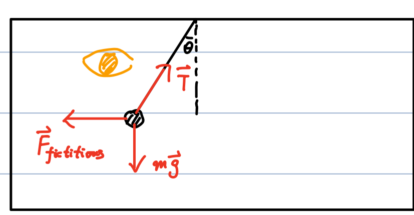


(ball remains static relative to the car)

$$\sum F_x = m a_x \Rightarrow T\sin\theta = m a_x \Rightarrow a_x = \frac{T}{m}\sin\theta$$

$$\sum F_y = 0 \Rightarrow T\cos\theta - mg = 0 \Rightarrow T = \frac{mg}{\cos\theta}$$

$$\Rightarrow a_x = g\tan\theta$$

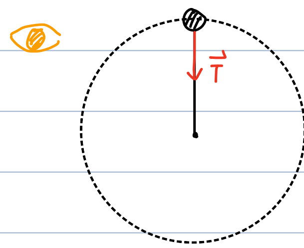


$$\sum F_x' = 0 \Rightarrow T\sin\theta - F_{fictitious} = 0$$

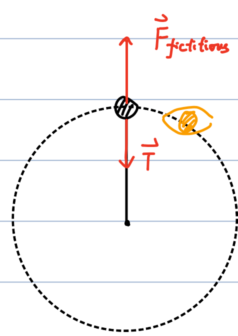
$$\sum F_y' = 0 \Rightarrow T\cos\theta - mg = 0 \Rightarrow T = \frac{mg}{\cos\theta}$$

$$\Rightarrow F_{fictitious} = T\sin\theta = mg\tan\theta = m a_x$$ (opposite direction)

Relative motion (non-inertial frame — uniform circular motion)



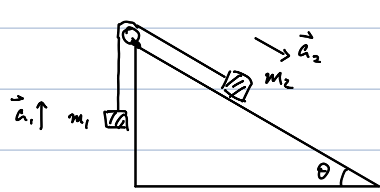
$$T = m\frac{v^2}{r}$$ (recall last class)



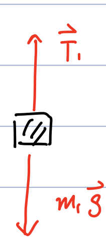
$$F_{fictitious} - T = 0$$

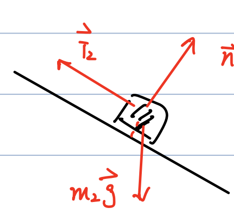
$$\Rightarrow F_{fictitious} = m\frac{v^2}{r} = ma$$ (opposite direction)

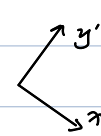
Example-1.



Free body diagram:







$$\Rightarrow \quad (T_1 - m_1 g)\,\hat{y} = m_1 \vec{a}_1 \qquad (m_2 g \sin\theta - T_2)\,\hat{x}' = m_2 \vec{a}_2$$

$$T_1 \to T, \quad \vec{a}_1 \to \vec{a}_1 = a\,\hat{y}, \quad T_2 \to T, \quad \vec{a}_2 \to \vec{a}_2 = a\,\hat{x}'$$

$$\Rightarrow \quad m_1 a + m_1 g = m_2 g \sin\theta - m_2 a$$

$$\Rightarrow \quad (m_1 + m_2)\,a = m_2 g \sin\theta - m_1 g$$

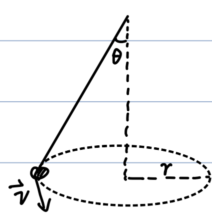
$$\Rightarrow \quad a = \frac{m_2 \sin\theta - m_1}{m_1 + m_2}\, g$$

$$\Rightarrow \quad T = m_1 (g + a)$$

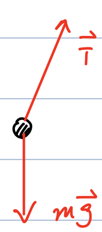
$$= m_1 g + m_1 \frac{m_2 \sin\theta - m_1}{m_1 + m_2}\, g$$

$$= \frac{m_1 m_2 (1 + \sin\theta)}{m_1 + m_2}\, g$$

Example- II Canonical Pendulum (圆锥摆)



r & $|\vec{v}|$ fixed



$$\begin{cases} T\cos\theta - mg = 0 & (\hat{y}) \\ T\sin\theta = m\dfrac{v^2}{r} & (\hat{x}) \end{cases}$$

$$\Rightarrow \quad \frac{mg}{\cos\theta} = \frac{mv^2}{r\sin\theta}$$

$$\Rightarrow \quad v = \sqrt{g r \tan\theta}$$

Drag (Fluid Resistance)

low speed: $\vec{R} = -b\vec{v}$

high speed: $\vec{R} = -\frac{1}{2}C\rho A v^2 \hat{v}$ $\qquad \hat{v} \equiv \frac{\vec{v}}{v}$

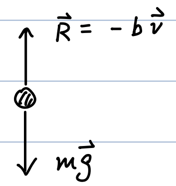
Notes: ① "$-$" : opposite to $\vec{v}$

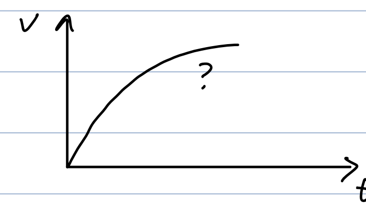
② $\vec{v}$: relative speed between object & fluid

③ ρ : density

A : effective cross-sectional area

- "Free"-fall again (assume b large, i.e. low speed)





$$mg - bv = ma, \quad a = \frac{dv}{dt}$$

$$\Rightarrow \quad g - \frac{b}{m}v = \frac{dv}{dt} \qquad v(t=0) = 0$$

$$\Rightarrow \quad v = \frac{mg}{b}\left(1 - e^{-\frac{b}{m}t}\right)$$

check:

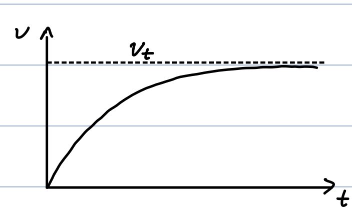
$$\dot{v} = \frac{mg}{b} e^{-\frac{b}{m}t} \frac{b}{m} = g e^{-\frac{b}{m}t}$$

$$\Rightarrow \dot{v} + \frac{b}{m}v = g e^{-\frac{b}{m}t} + g\left(1 - e^{-\frac{b}{m}t}\right) = g \quad ✓$$

Define $v_t \equiv \frac{mg}{b}$ terminal speed

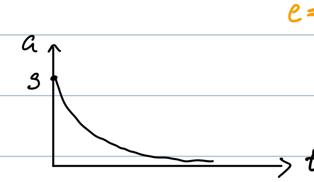
$\tau \equiv \frac{m}{b} = \frac{v_t}{g}$ characteristic time

$$\Rightarrow \quad v = v_t\left(1 - e^{-t/\tau}\right)$$

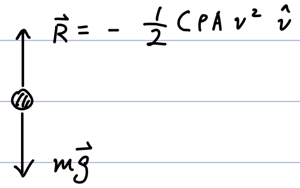


$$v(t=\tau) = v_t\left(1 - e^{-1}\right) \approx 0.63\, v_t \qquad e \approx 2.7183$$

acceleration: $a = \frac{dv}{dt} = g e^{-\frac{b}{m}t}$



Drag at high-speed



terminal speed:

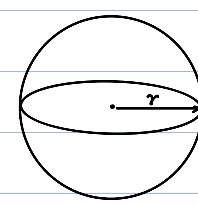
$$mg - \frac{1}{2} C \rho A v_t^2 \Rightarrow v_t = \sqrt{\frac{2mg}{C\rho A}}$$

evolution:

$$mg - \frac{1}{2} C \rho A v^2 = m \frac{dv}{dt}$$

$$\Rightarrow v = \sqrt{\frac{2mg}{C\rho A}} \tanh\left(\sqrt{\frac{gC\rho A}{2m}}\, t\right)$$

Q: v_t vs size of object?

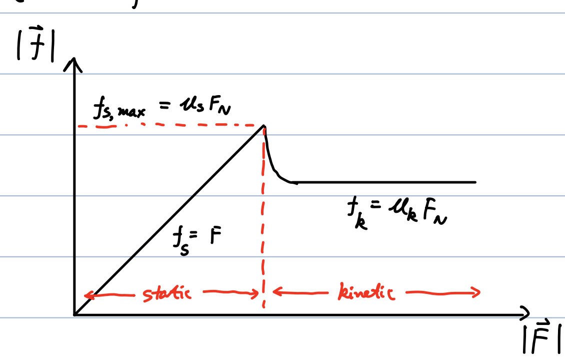


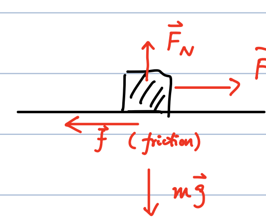
$$v_t = \sqrt{\frac{2mg}{C\rho A}}$$

$$\left.\begin{array}{l} m \propto r^3 \\ A \propto r^2 \end{array}\right\} \Rightarrow v_t \propto \sqrt{r}$$

减肥 ...

Frictional force



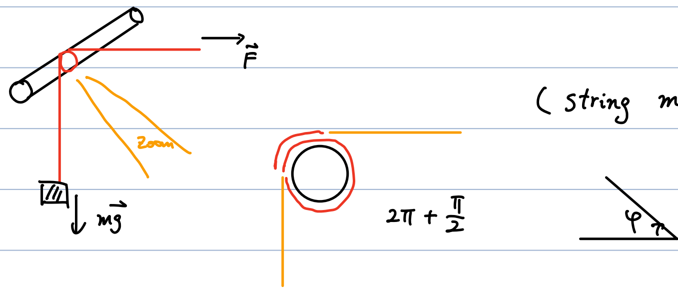


$\vec{F}_N$: normal force
(支持力、正压力)

μ_s: coefficient of static friction

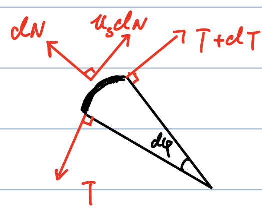
μ_k: - - - - - - kinetic friction

- Example . Static friction

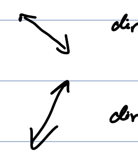


(string massless)

Lower limit (F small):



$$\begin{cases} dN = T \sin\frac{d\varphi}{2} + (T+dT)\sin\frac{d\varphi}{2} \\ (T+dT)\cos\frac{d\varphi}{2} + \mu_s dN = T\cos\frac{d\varphi}{2} \end{cases}$$



$$\Rightarrow \quad dN = T\,d\varphi$$

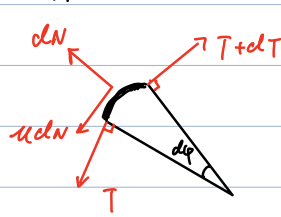
$$\Rightarrow \quad dT + \mu_s T\,d\varphi = 0$$

$$\Rightarrow \quad \int_{mg}^{F} \frac{dT}{T} = -\mu_s \int_0^{\frac{5}{2}\pi} d\varphi$$

$$\Rightarrow \quad \ln T \Big|_{mg}^{F} = -\mu_s \cdot \frac{5}{2}\pi$$

$$\Rightarrow \quad F = mg\, e^{-\frac{5}{2}\pi\mu_s}$$

Upper limit :



$$\Rightarrow \quad T + dT = T + \mu_s dN$$

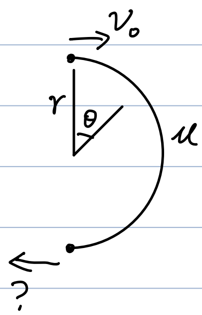
$$\Rightarrow \quad \int_{mg}^{F} \frac{dT}{T} = +\mu_s \int_0^{\frac{5}{2}\pi} d\varphi$$

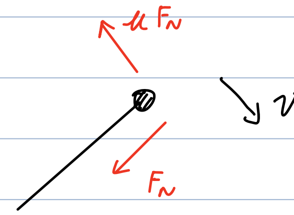
$$\Rightarrow \quad F = mg\, e^{+\frac{5}{2}\pi\mu_s}$$

In total :

$$mg\, e^{-\frac{5}{2}\pi\mu_s} \leq F \leq mg\, e^{\frac{5}{2}\pi\mu_s}$$

- Example . kinetic friction





$$\begin{cases} F_N = m\dfrac{v^2}{r} \\ -\mu F_N = m\dfrac{dv}{dt} \end{cases}$$

$$\Rightarrow \quad -\mu\frac{v^2}{r} = \frac{dv}{dt}$$

$$\Rightarrow \quad -\mu\frac{v}{r}\,dt = \frac{dv}{v}$$

$$\Rightarrow \quad -\frac{\mu}{r}\underbrace{\int_{t_0}^{t_f} v\,dt}_{\pi r} = \underbrace{\int_{v_0}^{v_f}\frac{dv}{v}}_{\ln v_f - \ln v_0}$$

$$\Rightarrow \quad v_f = v_0\, e^{-\mu\pi}$$